THREE PHYLOGENETIC TREES ON FOURTEEN LEAVES ALWAYS SHARE A QUARTET

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Abstract. Snir and Yuster asked for $h(k)$, the least n such that any k unrooted binary phylogenetic trees on the same n leaves resolve some four leaves identically. It was known that $h(2) = 6$ and $11 \le h(3) \le 14$; for general k the upper bound was a four-fold iterated exponential. We prove $h(3) = 14$. The lower bound is a tree T on thirteen leaves and a leaf permutation σ of order three such that T, σT and $\sigma^2 T$ have no common quartet. For the upper bound we give an exact finite version of the Snir–Yuster argument: an exhaustive check over the 135 tree shapes on fourteen leaves shows that a balanced edge of the first tree and a suitable edge of the second always leave separated leaf sets of sizes three and four, and a SAT computation with DRAT-checked certificates confirms the two-tree statement independently. Iterating the underlying balanced-cut step, balanced on the larger set and with the potential $\min\{|X||Y|, 2|X|^2, 2|Y|^2\}$, gives the elementary bound $h(k) \le \lfloor 3\beta^{k-1} \rfloor + 1$ for all k, where $\beta = (1+\sqrt{13})/2$; done exactly by dynamic programming, the same bookkeeping gives $h(4) \le 28$, $h(5) \le 61$ and $h(6) \le 125$. Together with the caterpillar families of Snir and Yuster, for which we give an explicit construction, this gives $2^k + 2 \le h(k) \le \lfloor 3\beta^{k-1} \rfloor + 1$. A symmetric SAT construction gives $h(4) \ge 23$, so $23 \le h(4) \le 28$.

1. Introduction

A *phylogenetic tree* on $[n] = \{1, \dots, n\}$ is an unrooted tree whose internal vertices have degree three and whose leaves are labelled bijectively by $[n]$. Restricting a tree to a 4-set Q and suppressing degree-two vertices leaves one internal edge, which splits Q into two pairs: the *quartet split* $ab|cd$, $ac|bd$ or $ad|bc$. A family of trees on $[n]$ has a *common quartet* if some 4-set has the same split in every member. Let $h(k)$ be the least n such that every k trees on $[n]$ have a common quartet, and put $N(k) = h(k) - 1$. Restricting all trees to a subset of the leaves preserves the absence of a common quartet, so k trees without one exist on n leaves exactly when $n \le N(k)$. Clearly $h(1) = 4$ and $N(k)$ is nondecreasing in k.

Bandelt and Dress [1] noted that two trees on at least six leaves share a quartet, so $h(2) = 6$. Snir and Yuster [7, Sec. 5] introduced $h(k)$ and its caterpillar analogue $h^*(k)$, for which they state $h^*(k) = 2^k + 2$, remarked that determining $h(k)$ "seems considerably more involved", and showed $h(3) > 10$ with three trees on ten leaves. Since caterpillars are phylogenetic trees, $h(k) \ge h^*(k)$; Snir and Yuster omit the construction behind $h^*(k) \ge 2^k + 2$, and we give one in Section 4. Their theorem that any three trees on n leaves have separated leaf sets of size at least $2n/27$ [7, Thm. 1.3] gives $h(3) \le 14$, as Currie, Czabarka, Fischer, Simmons, Székely and Wicke observed [5], who also proved

$$8^{1/4} 3^{k/4} \le h(k) \le \mathrm{tw}_3\big(6(k-1)2^k + O(\log k)\big)$$

through hypergraph Ramsey numbers [5, Thm. 5].

Theorem 1.1. $h(3) = 14$.

Theorem 1.2. *Put* $\beta = (1 + \sqrt{13})/2$. *Then* $2^k + 2 \le h(k) \le \lfloor 3\beta^{k-1} \rfloor + 1$ *for every* $k \ge 1$.

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Theorem 1.2 replaces the iterated exponential by a single exponential and pins the growth rate between 2 and $\beta \approx 2.3028$: $\log 2 \le \liminf_k \frac{1}{k}\log h(k) \le \limsup_k \frac{1}{k}\log h(k) \le \log \beta$. The lower bound is the caterpillar bound of Snir and Yuster; Currie et al. state the weaker $h(k) \ge 8^{1/4}3^{k/4}$. The upper bound iterates the balanced-cut and pigeonhole steps that Snir and Yuster use for two and three trees (their Proposition 2.4 and the proof of Theorem 2.5) and for paths (their Proposition 5.2). The plain iteration gives $h(k) \le 2 \cdot 3^{k-1} + 1$ (Proposition 2.3); what is new is the iteration for trees with exact bookkeeping, and the reduction of the base from 3 to β by balancing the cut on the larger set and tracking a potential instead of the smaller set (Lemma 2.6). Done exactly, by dynamic programming over coloured trees, the same bookkeeping gives $h(4) \le 28$, $h(5) \le 61$ and $h(6) \le 125$ (Theorem 2.9). Table 1 collects small values.

k	lower bound	source	upper bound	source
1	4	trivial	4	trivial
2	6	[1]	6	[1]
3	14	Proposition 3.1	14	Theorem 1.1
4	23	SAT, σ of type $4^5 2$	28	Theorem 2.9
5	34	caterpillars, Proposition 4.1	61	Theorem 2.9
6	66	caterpillars, Proposition 4.1	125	Theorem 2.9
k	$2^k + 2$	caterpillars, Proposition 4.1	$\lfloor 3\beta^{k-1} \rfloor + 1$	Theorem 2.7

TABLE 1. Bounds on $h(k)$ for small k. Before this note the bounds for $k = 3$ were 11 and 14, and for $k \ge 4$ the upper bounds were iterated exponentials. For $k = 4$ the caterpillar bound is 18.

2. Separated sets

Disjoint leaf sets X and Y are *separated* in a tree if some edge has all of X on one side and all of Y on the other. If X and Y are separated in every tree of a family and $x, x' \in X$, $y, y' \in Y$ are distinct, every tree resolves $\{x, x', y, y'\}$ as $xx'|yy'$, a common quartet.

Lemma 2.1 (balanced edge)**.** *Every phylogenetic tree with $m \ge 2$ leaves has an edge with at least $m/3$ leaves on each side.*

Proof. This is classical; we include the short argument. Choose an edge uv minimising the larger number M of leaves on one side, and let that side contain v. If $M > 2m/3$ then $M \ge 2$, so v is internal and its two other edges cut the M leaves into parts $a + b = M$ with $a, b \ge 1$. The edge leading to the a leaves has sides $a < M$ and $m - a$, so minimality gives $m - a \ge M$; likewise $m - b \ge M$. Adding, $2m - M \ge 2M$, a contradiction. □

For a set S of at least two leaves of T, the restriction $T|S$ is the tree spanned by S with degree-two vertices suppressed. Each edge of $T|S$ comes from a path of T, and removing any edge of that path splits S as the edge of $T|S$ does. So Lemma 2.1 gives an edge of T splitting S into two parts with at least $|S|/3$ elements each.

Lemma 2.2 (step)**.** *Let X, Y be separated in trees $T_1, \dots, T_j$, and let an edge of a further tree T' split $X \cup Y$ into W and Z. If $|X|, |Y|, |W|, |Z| \ge 2t-1$, then one of the pairs $(X \cap Z,\, Y \cap W)$ and $(X \cap W,\, Y \cap Z)$ consists of two sets with at least t elements each. Both pairs are separated in $T_1, \dots, T_j, T'$.*

Proof. Put $A = X \cap Z$, $B = X \cap W$, $C = Y \cap Z$ and $D = Y \cap W$. Subsets of X and Y remain separated in each T_i, and $A, C \subseteq Z$, $B, D \subseteq W$ are separated in T'. If $\min\{|A|, |D|\} \le t - 1$ and $\min\{|B|, |C|\} \le t - 1$, then one of $|A| + |B| = |X|$, $|A| + |C| = |Z|$, $|D| + |B| = |W|$, $|D| + |C| = |Y|$ is at most $2t - 2$. □

Proposition 2.3. *$h(k) \le 2 \cdot 3^{k-1} + 1$ for every $k \ge 2$.*

Proof. Let $\phi(s) = \lfloor (\lceil 2s/3 \rceil + 1)/2 \rfloor$. If X, Y are separated in some trees and $|X|, |Y| \ge s$, Lemma 2.1 applied to $T'|(X \cup Y)$ gives $|W|, |Z| \ge \lceil 2s/3 \rceil$, and $\lceil 2s/3 \rceil \le s$, so Lemma 2.2 with $t = \phi(s)$ produces a pair of size at least $\phi(s)$ separated in one more tree. Moreover $\phi(s) \ge a$ if and only if $\lceil 2s/3 \rceil \ge 2a - 1$, that is $s \ge 3a - 2$. Define $b_1 = 2$ and $b_{i+1} = 3b_i - 2$, so $b_i = 3^{i-1} + 1$: a pair of size b_i separated in some trees can be pushed through $i - 1$ further trees and still have size at least 2.

Let $T_1, \dots, T_k$ be trees on $n \ge 2 \cdot 3^{k-1} + 1$ leaves, $k \ge 2$. Balanced edges of T_1 and of T_2 split $[n]$ into $X \cup Y$ and $W \cup Z$ with all four parts of size at least $\lceil n/3 \rceil \ge 2 \cdot 3^{k-2} + 1$. Lemma 2.2 with $t = 3^{k-2} + 1 = b_{k-1}$ gives a pair of size b_{k-1} separated in T_1, T_2. Pushing it through $T_3, \dots, T_k$ yields two sets of size at least 2 separated in all k trees, hence a common quartet. □

Remark 2.4. For $k = 2$ the bound is 7 (the truth is 6), and for $k = 3$ it is 19. The executable file `iterate_check.py` runs this proof on random families at the threshold $n = 2 \cdot 3^{k-1} + 1$ for $k \le 5$ and checks every intermediate size bound; it also runs the proof of Theorem 2.7 below for $k \le 6$, checking the potential with exact rationals.

A smaller base. Proposition 2.3 loses a factor 3 per tree because it only tracks the smaller of the two sets. Balancing the cut on the larger set and tracking a potential does better.

Lemma 2.5 (weighted balanced edge)**.** *Let T be a phylogenetic tree with at least two leaves, and give the leaves nonnegative weights with total 1, each at most $1/2$. Then some edge of T has weight at least $1/3$ on each side.*

Proof. If some edge has weight at most $1/2$ on each side, both sides weigh $1/2$. Otherwise direct every edge towards its side of weight more than $1/2$. Two components of $T - v$ cannot both weigh more than $1/2$, so every vertex has at most one outgoing edge; as a tree has more vertices than edges, some vertex v has none, so every component of $T - v$ weighs at most $1/2$. A leaf v would then be a side of weight more than $1/2$ by itself, so v is internal and its three components have total weight 1. The heaviest weighs between $1/3$ and $1/2$, and the edge joining it to v has weight at least $1/3$ on each side. □

For real $a, d \ge 0$ put $\Psi(a, d) = \min\{ad,\ 2a^2,\ 2d^2\}$. It is symmetric, nondecreasing in each argument and homogeneous of degree two, and $\Psi(a, d) \le 2 \min\{a, d\}^2$.

Lemma 2.6 (potential step)**.** *Put $t = (\sqrt{13} - 1)/6$ and $c = t^2 = (7 - \sqrt{13})/18$. Let $0 < a \le d$, $x \in [0, 1]$ and $y \in [1/3, 2/3]$. Then*

$$\max\big\{\Psi(xa, (1-y)d),\ \Psi((1-x)a, yd)\big\} \ \ge\ c\,\Psi(a, d).$$

The constant c is best possible for this real-variable inequality.

Proof. The constants satisfy $3t^2 + t = 1$, $0 < t < 1/2$ and $c < 2/9$. Call the two terms O_1 and O_2. Replacing (x, y) by $(1 - x, 1 - y)$ exchanges them and preserves the hypothesis, so we may assume $x + y \le 1$; by homogeneity $a = 1$ and $d = r \ge 1$. If $r > 2$ then $\Psi(1, r) = \Psi(1, 2) = 2$ while O_1, O_2 only decrease when r is lowered to 2, so we may assume $1 \le r \le 2$, and the target is $c\Psi(1, r) = cr$.

Case $x \ge t$. Since $1 - y \ge x$, the three entries of O_1 satisfy

$$x(1-y)r \ge x^2 r \ge cr, \qquad 2x^2 \ge 2c \ge cr, \qquad 2(1-y)^2 r^2 \ge 2cr^2 \ge cr.$$

Case $x < t$. The three entries of O_2 satisfy

$$(1-x)yr \ge (1-t)r/3 = cr, \qquad 2(1-x)^2 \ge 2(1-t)^2 \ge 2c \ge cr,$$

and $2y^2 r^2 \ge 2r^2/9 \ge cr$. Thus one output is at least cr in either case.

For sharpness, take $a = 1$, $d = 2$, $x = t$, $y = 1/3$. The input potential is 2, while both outputs equal $2c$: $\Psi(t, 4/3) = 2t^2$ and $\Psi(1 - t, 2/3) = 2(1 - t)/3$. This equality example is in the real relaxation; its irrational x is not a ratio of finite leaf counts. □

The sharp inequality and its equality example are checked in Lean 4 with Mathlib, in the file `CommonQuartetPotential.lean`. The two theorems are

`potential_step_sharp`,

`potential_step_sharp_witness`.

They use no `sorry`; the axiom audit reports only `propext`, `Classical.choice` and `Quot.sound`. The earlier theorem `potential_step` retains the weaker factor $3/16$. Only these real-variable claims are formalised: the tree lemmas, iteration, dynamic programme and value $h(3) = 14$ are not Lean theorems in the supplied file.

Theorem 2.7. *If $n > 3\beta^{k-1}$, where $\beta = (1+\sqrt{13})/2 = 1/t$, any k phylogenetic trees on $[n]$ have a common quartet. Hence $h(k) \le \lfloor 3\beta^{k-1} \rfloor + 1$ for every $k \ge 1$; for $k = 1, \dots, 6$ this is $4, 7, 16, 37, 85, 195$.*

Proof. Lemma 2.5 with weights $1/n$ gives an edge of T_1 splitting $[n]$ into X, Y with $n/3 \le |X| \le n/2 \le |Y|$ (up to swapping), so $\Psi(|X|,|Y|) \ge \min\{\frac{n}{3} \cdot \frac{2n}{3}, \frac{2n^2}{9}\} = 2n^2/9$. Suppose X, Y are separated in $T_1, \dots, T_j$, with $a = |X| \le d = |Y|$ and $d \ge 2$. Give the leaves of $T_{j+1}|(X \cup Y)$ weight 0 on X and $1/d$ on Y. Lemma 2.5 and the remark after Lemma 2.1 give an edge of T_{j+1} splitting $X \cup Y$ into Z and W with $y = |Y \cap Z|/d \in [1/3, 2/3]$; put $x = |X \cap Z|/a$. As in Lemma 2.2, both pairs $(X \cap Z, Y \cap W)$ and $(X \cap W, Y \cap Z)$ are separated in $T_1, \dots, T_{j+1}$, and by Lemma 2.6 one of them has potential at least $c\Psi(a,d)$. Relabel the retained pair so that the smaller set is X before the next step; symmetry of Ψ preserves its value. After k trees the potential is at least $\frac{2n^2}{9}c^{k-1} > 2$; it therefore exceeds 2 at every stage, which keeps both sets of size at least two throughout, since $\Psi \le 2\min\{|X|,|Y|\}^2$. The two final sets give a common quartet. □

Theorem 1.2 follows from Theorem 2.7 and Proposition 4.1 below. Theorem 2.7 is better than Proposition 2.3 for every $k \ge 3$, but for $k = 2, 3, 4, 5$ its bounds $7, 16, 37, 85$ are weaker than the available $6, 14, 28, 61$. The thresholds are computed exactly using $\beta^2 = \beta + 3$; at $k = 3$ the condition is $n > (21 + 3\sqrt{13})/2$, hence $n \ge 16$. Equal total weight on X and Y instead gives the factor $4/25$ (tight at $d = 2a$, $x = 2/5$, $y = 4/15$) and base $5/2$. Lemma 2.6 proves the exact best factor $c \approx 0.18858$ for this potential and this cut condition over the reals; it does not establish optimality among all tree strategies or potentials. A numerical solution of the underlying game, in which a separated pair is pushed through one tree at a time with any choice of weights, suggests that no strategy of this kind beats a base of about $3/\sqrt{2} \approx 2.12$. That numerical statement remains unproved.

Exact bookkeeping for small k. For small k the one-pair strategy can be optimised exactly. Call a pair of sizes (u, v) 0-*safe* if $\min\{u, v\} \ge 2$. For $m \ge 1$, colour u leaves red and v leaves blue; an edge of a tree on these $u + v$ leaves with i red and j blue leaves on one side is *good* if $(i, v - j)$ or $(u - i, j)$ dominates an $(m-1)$-safe pair (componentwise, after possibly swapping its entries), and (u, v) is m-*safe* if it dominates a pair (u', v') such that every tree with u' red and v' blue leaves has a good edge.

Lemma 2.8. *Let X, Y be separated in $T_1, \dots, T_j$, with $(|X|,|Y|)$ m-safe. Then any further trees $T_{j+1}, \dots, T_{j+m}$ have a common quartet with $T_1, \dots, T_j$. Consequently, if every tree on n leaves has an edge splitting $[n]$ into parts of sizes s and $n - s$ with $(s, n-s)$ $(k-1)$-safe, then $h(k) \le n$.*

Proof. Shrink X, Y to sizes (u', v') as in the definition and colour them. The restriction $T_{j+1}|(X \cup Y)$ has a good edge, which comes from an edge of T_{j+1}; as in Lemma 2.2 it yields a pair separated in $T_1, \dots, T_{j+1}$ whose sizes dominate an $(m-1)$-safe pair. Induct on m; at $m = 0$ two sets of size at least two give a common quartet. □

Whether some tree with u red and v blue leaves has no good edge is decided by dynamic programming. Root such a tree at an internal vertex; every edge is then the top edge of a unique rooted subtree, and the edge is bad exactly when the colour counts (i, j) of that subtree are bad. Call (i, j) *realisable* if it is bad and is $(1, 0)$ or $(0, 1)$ or the sum of two realisable pairs. A tree without good edges exists if and only if (u, v) is the sum of three realisable pairs. The same recursion with one colour decides the condition on the first tree in Lemma 2.8.

Theorem 2.9 (computer-assisted)**.** *$h(4) \le 28$, $h(5) \le 61$ and $h(6) \le 125$. The same computation gives $h(2) \le 7$ and $h(3) \le 14$.*

Proof. For $k = 4$ the needed safe pairs are $(3, 4)$ (1-safe), $(5, 9)$ and $(6, 7)$ (2-safe) and $(10, 18)$, $(11, 16)$, $(13, 14)$ (3-safe). By Lemma 2.1 every tree on 28 leaves has an edge with $s \in \{10, \dots, 14\}$ leaves on its smaller side, and $(s, 28 - s)$ dominates one of the 3-safe pairs. For $k = 5$ six further 4-safe pairs between $(20, 41)$ and $(25, 31)$, and $(9, 22)$ at level 3, are used, and for $k = 6$ eighteen further pairs. Each claim is one run of the recursion above, done twice by independent programs (Section 5). □

The recursion supplies certified bounds by pushing one separated pair through the trees one at a time. The resulting bound is sharp for $k = 3$ but not for $k = 2$, where $h(2) = 6$, and the successive ratios $1.75, 2, 2, 2.18, 2.05$ of its values $4, 7, 14, 28, 61, 125$ are in line with the base $3/\sqrt{2} \approx 2.12$ suggested above.

3. Three trees

Thirteen leaves. Let $\sigma = (1\,2\,3)(4\,5\,6)(7\,8\,9)(10\,11\,12)$, fixing 13, and let

$$T = (1, (3, 6), (((7, 9), (4, (5, 12))), ((10, 11), (2, (8, 13)))))$$

in Newick notation. Relabelling by σ and σ^2 gives

$$\sigma T = (2, (1, 4), (((8, 7), (5, (6, 10))), ((11, 12), (3, (9, 13))))),$$
$$\sigma^2 T = (3, (2, 5), (((9, 8), (6, (4, 11))), ((12, 10), (1, (7, 13))))).$$

Proposition 3.1. *No 4-subset of $[13]$ has the same split in T, σT and $\sigma^2 T$. Hence $h(3) \ge 14$.*

Proof. This is a check of $\binom{13}{4} = 715$ subsets. In each tree the split of $\{a, b, c, d\}$ is the pairing minimising $\mathrm{d}(a, b) + \mathrm{d}(c, d)$, $\mathrm{d}(a, c) + \mathrm{d}(b, d)$, $\mathrm{d}(a, d) + \mathrm{d}(b, c)$, with d the path length (the four-point condition). A standard-library script finds no 4-set with three equal splits; each pair of the three trees shares exactly 137 quartets. □

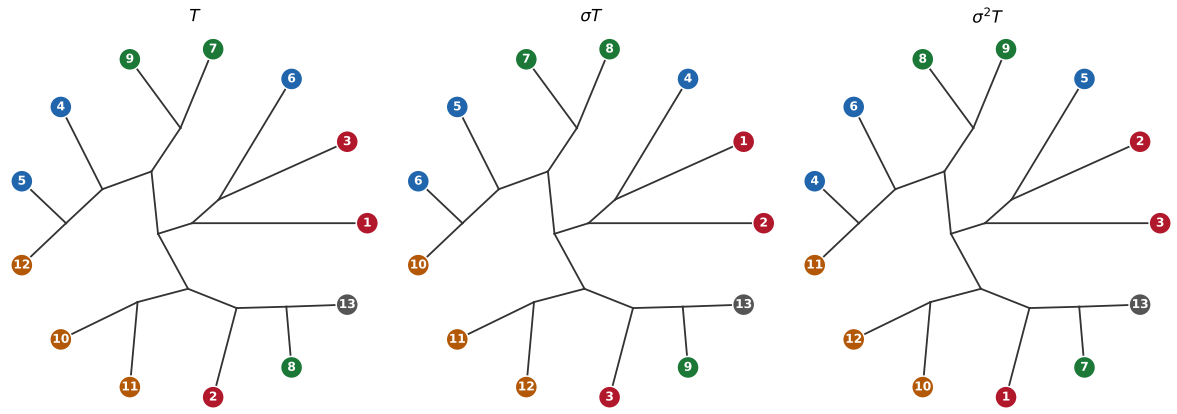


Figure 1. The trees T, σT, $\sigma^2 T$ of Proposition 3.1 in one layout. Colours mark the σ-orbits $\{1, 2, 3\}$, $\{4, 5, 6\}$, $\{7, 8, 9\}$, $\{10, 11, 12\}$ and $\{13\}$. No four leaves are split the same way in all three drawings.

The tree was found by a SAT search in which T is the only free tree (Section 5). On thirteen leaves the cycle types $3^4 1$ and $6^2 1$ of σ admit solutions, while 13, $12\,1$, $5^2 3$ and $4^3 1$ do not. On fourteen leaves the types $3^4 1^2$, $3^4 2$, $6^2 1^2$ and 14 admit none, in line with Theorem 1.1.

Fourteen leaves. The upper bound $h(3) \le 14$ is the instance $n = 14$ of Snir and Yuster's three-tree separation theorem, whose most technical ingredient is a two-tree lemma with set sizes $4n/27$ and $4n/9$ [7, Lemma 2.6], proved with real weights and then stated in an error-free form in their Section 4. At $n = 14$ that error-free form asserts exactly Lemma 3.3 below. We prove a stronger statement, Lemma 3.4, by an exhaustive check that takes about a minute, and Lemma 3.3 separately by a certified SAT computation, so that $h(3) \le 14$ rests on an exact finite argument checked in two independent ways.

Lemma 3.2. *Let X, Y be disjoint leaf sets of a tree T with $|X|, |Y| \ge 2$. They are separated in T if and only if T resolves $xx'|yy'$ for all distinct $x, x' \in X$ and distinct $y, y' \in Y$.*

Proof. One direction was noted above. If X and Y are not separated, the subtrees spanned by X and by Y share a vertex v, which is internal. Let C_1, C_2, C_3 be the components of $T - v$; each of X and Y meets at least two of them. Take $x \in X \cap C_1$ and $x' \in X \cap C_2$. If Y meets C_3, take $y \in Y \cap C_3$ and $y' \in Y \cap (C_1 \cup C_2)$; then y' is paired with x or x'. Otherwise take $y \in Y \cap C_1$ and $y' \in Y \cap C_2$, giving $xy|x'y'$. Either way $xx'|yy'$ fails. □

Lemma 3.3 (computer-assisted, Section 5)**.** *Any two phylogenetic trees on* [14] *have disjoint leaf sets X, Y with $\{|X|, |Y|\} = \{3, 4\}$ separated in both.*

Lemma 3.4 (exhaustive check)**.** *Let $[14] = X \cup Y$ with $X \cap Y = \emptyset$ and $|X|, |Y| \ge 5$, and let T be a tree on* [14]*. Then some edge of T splits* [14] *into Z and W such that one of the pairs $(X \cap Z, Y \cap W)$ and $(X \cap W, Y \cap Z)$ consists of a set of at least three and a set of at least four elements. On* 13 *leaves the same statement holds when $\{|X|, |Y|\} = \{6, 7\}$ and fails when $\{|X|, |Y|\} = \{5, 8\}$.*

Proof. Relabelling is free, so it suffices to take every unrooted shape on 14 leaves (there are 135) and every choice of 5, 6 or 7 of its leaves as X, and to inspect all edges: $135 \cdot (2002 + 3003 + 3432) =$ 1,138,995 cases, none without such an edge. On 13 leaves the 66 shapes give no failure for $|X| = 6$ and 891 failures for $|X| = 5$. The script `bipartition_check.py` uses only the standard library. □

Proof of Theorem 1.1. The lower bound is Proposition 3.1. Given three trees on 14 leaves, Lemma 2.1 gives an edge of T_1 splitting [14] into X, Y with $|X|, |Y| \ge 5$, and Lemma 3.4 applied to T_2 gives sets $X' \subseteq X$ and $Y' \subseteq Y$ separated in T_1 and T_2 with $\{|X'|, |Y'|\} = \{3, 4\}$ after shrinking (Lemma 3.3 gives such sets directly). Lemma 2.1 applied to $T_3|(X' \cup Y')$ gives an edge of T_3 splitting these seven leaves into W, Z with $|W|, |Z| \ge 3$, and Lemma 2.2 with $t = 2$ applied to X', Y' gives two sets of at least two leaves separated in all three trees. □

Lemma 3.3 is sharp: by Proposition 3.1 and the proof above it fails on thirteen leaves, and a brute-force search confirms that no two of $T, \sigma T, \sigma^2 T$ have separated sets of sizes three and four.

4. Lower bounds for general k

A linear order of leaves specifies a caterpillar (reversal and swaps within the two end cherries give the same tree), and it splits a 4-set into the two leaves that come first and the two that come last. Let $L = 2^{k-1}$ and $m = 2L + 1$. For $1 \le j \le L$ call $\{j,\, m + 1 - j\}$ the j-th layer, and call $L + 1$ the centre. For $t = 0, \dots, k - 1$ let C_t be the caterpillar whose order has, at position j, the element j if $\beta_t(j) = 0$ and $m + 1 - j$ if $\beta_t(j) = 1$, the centre at position $L + 1$, and the other element of layer j at position $m + 1 - j$. Here $\beta_0 \equiv 0$ and $\beta_t(j)$ is binary digit $t - 1$ of $j - 1$; thus C_t flips the layers j whose digit $t - 1$ of $j - 1$ is one.

Proposition 4.1. *The caterpillars $C_0, \dots, C_{k-1}$ on $2^k + 1$ leaves have no common quartet. Hence $h(k) \geq h^*(k) \geq 2^k + 2$.*

Proof. Induction on k; for $k = 1$ there are three leaves. In every order each layer occupies positions j and $m + 1 - j$, so the elements of the outer layers $1, \dots, L/2$ always occupy the first and last $L/2$ positions, and the inner layers and the centre always occupy the middle block. In C_{k-1} every inner layer is flipped and no outer layer is, so its order is that of C_0 with the middle block reversed. Restricted to the inner layers and the centre, and also restricted to the outer layers and the centre, the orders of $C_0, \dots, C_{k-2}$ form the family for $k - 1$ after relabelling, since only digit $k - 2$ of $j - 1$ distinguishes the two halves. Let Q be a 4-set.

If Q lies in the inner block, or in the outer layers together with the centre, induction applies. If Q has three outer elements and one inner element x, replacing x by the centre changes no split, because both lie in the middle block in every order. If Q has one outer element o, then o is first or last in every order and pairs with the inner element nearest to it; this element differs between C_0 and C_{k-1}, since the middle block is reversed and Q has three inner elements. If Q has two outer elements o_1, o_2 and two inner elements, then when o_1 and o_2 are on opposite sides they pair with different inner elements in C_0 and C_{k-1}. Otherwise they are on the same side in C_0, so they lie in different layers $j_1 \neq j_2$ and are both the smaller or both the larger element of their layers; choose a digit in which $j_1 - 1$ and $j_2 - 1$ differ, so that exactly one of the two layers is flipped in the corresponding C_t. In C_t the elements o_1, o_2 are on opposite sides, so C_t splits Q differently from C_0. At $k = 2$ there is only one outer layer, so this same-side subcase is empty. □

For $k = 3$ the orders are 123456789, 183654729 and 127654389; Snir and Yuster [7, Sec. 5] list the same three caterpillars. A standard-library script checks Proposition 4.1 directly for $k \leq 6$. For $k = 4$ the bound 18 is weaker than the 23 from four trees on 22 leaves found by a symmetric SAT search (Section 5).

5. Computations

Shapes. Relabelling leaves preserves both properties studied, so the first tree may be a fixed representative of its unlabelled shape. Deleting a leaf of an unrooted shape and rooting at its former neighbour gives a rooted binary shape on one leaf fewer, so the shapes with n leaves come from the rooted shapes with $n - 1$ leaves, deduplicated by a canonical form. The counts $2, 2, 4, 6, 11, 18, 37, 66, 135$ for $n = 6, \dots, 14$ agree with OEIS A000672 [6], and an independent enumeration by leaf insertion gives the same counts.

Free trees. A free tree is encoded by three variables per 4-set $\{a < b < c < d\}$, for the splits $ab|cd$, $ac|bd$ and $ad|bc$, with an exactly-one constraint; for every 5-set the five chosen splits must form the quartet system of one of the 15 binary trees on those five leaves. Every tree satisfies these constraints, so an unsatisfiable formula excludes all trees. By a theorem of Colonius and Schulze [4] and Bandelt and Dress [1] the constraints characterise tree quartet systems, but the arguments never rely on this: every satisfying assignment reported here was rebuilt into a tree by leaf insertion and checked directly.

Lemma 3.3. With the first tree T_1 fixed, add for every edge split $P|Q$ of T_1 and all $X \subseteq P$, $Y \subseteq Q$ (or the reverse) with $\{|X|, |Y|\} = \{3, 4\}$ a clause stating that one of the 18 cross quartets $xx'|yy'$ is not chosen in the free tree T_2. By Lemma 3.2 the models are exactly the trees T_2 that violate the lemma together with T_1. All 135 formulas, with 33,033 variables and between 160,668 and 163,020 clauses, are unsatisfiable. Each was solved by CaDiCaL 3.0.1 [3] with a DRAT proof, and every proof was verified by DRAT-trim [8]; the proofs total 3.2 GB and the solving took 2.7 hours of wall time on a shared machine. The CNF files are regenerated deterministically and their SHA-256 hashes are recorded. A second implementation sharing no code with the first generates the five-leaf patterns from the 15 labelled trees, enumerates shapes by leaf insertion, forbids the 228 non-tree patterns by direct clauses and is solved by Kissat 4.0.4 [2]; it also reports all 135 formulas unsatisfiable. As a positive control, the same

encoding on thirteen leaves with T_1 of the shape of T is satisfiable, and the decoded second tree has no separated pair of sizes three and four.

Theorem 2.9. One program (with numpy) computes, level by level, the least v for which (u, v) is m-safe, for u up to a bound, and the resulting bounds on $h(k)$. A second program, written separately with the standard library only, reads the claimed table and re-derives every claim used by the stated bounds with its own implementation of the recursion; it rejects the claims $h(4) \le 27$ and $h(3) \le 14$ when the claimed tables are lowered by one. The recursion was compared with brute-force enumeration of all labelled trees on up to eight leaves for about 1000 random sets of bad colour counts, with full agreement, and its level-two values $(5, 9)$ and $(6, 7)$ agree with the exhaustive check of Lemma 3.4.

Searches. For k trees, fixing T_1 and leaving $T_2, \dots, T_k$ free, one clause per 4-set forbids all free trees from taking the split of T_1. In the symmetric variant T is the only free tree, the family is $T, \sigma T, \dots, \sigma^{k-1}T$, and for each 4-set Q and pairing s a clause forbids T from resolving every $\sigma^{-i}Q$ by $\sigma^{-i}s$. As sanity checks the method reproduces $h(2) = 6$ and the caterpillar value $h^*(3) = 10$ of [7], for which all trees are also forbidden to contain three cherries on six leaves. The construction for $k = 4$ in Table 1 comes from a symmetric search with σ of cycle type $4^5 2$ on 22 leaves and was verified with the standard-library checker; for $k = 5$ the same method reached only 27 leaves, fewer than the 33 of Proposition 4.1. On twelve leaves the three-tree formulas with T_1 among the first five shapes, including the caterpillar, are unsatisfiable with checked DRAT proofs; thus three trees on twelve leaves, one a caterpillar, always share a quartet, while such triples exist on eleven leaves.

6. Questions

The value of $h(4)$ lies between 23 and 28. The certified one-pair strategy of Theorem 2.9 gives 28. An exploratory variant starting from three branches at a vertex also gave 28. The supplied verifier establishes the safe pairs used by our upper bounds; it does not certify global optimality among all one-pair strategies. The exponential growth rate of $h(k)$ lies between 2 and β. Since $h(3) = 14 > h^*(3)$ and $h(4) \ge 23 > h^*(4)$, general trees do better than caterpillars for small k; is $h(k) = O(2^k)$? On the upper side, the numerical value $3/\sqrt{2}$ of the one-pair game suggests that reaching base 2 needs an argument that tracks more than one separated pair. It would also be interesting to know whether in every triple on thirteen leaves without a common quartet the three trees have the same shape; on twelve leaves there are triples in which they do not.

Data and code

The trees, checkers, CNF generators, solver logs and hashes are in the `verification` directory next to this paper. The Newick checker, the separation checker, the proof executor, the exhaustive check of Lemma 3.4, the checker for Theorem 2.9, the caterpillar check and both CNF generators for Lemma 3.3 use only the Python standard library.

Contributions and AI use

Harvey Yorke directed the project. Coding agents developed the arguments, the verification checkers and the manuscript drafts. Valyu DeepResearch carried out the literature, priority and internal adversarial reviews. These reviews are internal; the manuscript has not undergone external peer review. Harvey Yorke has not yet completed a line-by-line review of every proof; correctness rests on the written proofs and on the checks and any formal verification described in this paper.

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